

KOM4531 - Omron Sysmac Competence Laboratory

In this tutorial, a guidance for laboratory sets in Omron Sysmac Competence Laboratory is given. You will learn how to create new project, add *EtherCAT*¹ devices, add coupler units, configure *I/O Map* and write Ladder diagrams.

1 Laboratory Set

Every set is equipped with Windows 7 OS which has **Sysmac Studio** program and **NJ301-1100 CPU V1.09**. These devices are connected via an ethernet hub. The CPU has 3 EtherCAT slaves of 2 **R88D-KN01H-ECT G5 Series ServoDrive/Motor V2.1** and **NX-ECC202 EtherCAT coupler V1.2**. The coupler has units given in Table 1.

Model	Product	Revision
NX-SL3300	Safety CPU Unit	1.0
NX-SIH400	Safety Input Unit	1.0
NX-SIH400	Safety Input Unit	1.0
NX-SOD400	Safety Output Unit	1.0
NX-ID3444	DC Input Unit	1.0
NX-ID4442	DC Input Unit	1.0
NX-OD2258	Transistor Output Unit	1.0
NX-OD4256	Transistor Output Unit	1.0
NX-AD2603	Analog Input Unit	1.0
NX-DA3603	Analog Output Unit	1.0

Table 1: Units of the coupler

1st servo drive is connected to a rotary plant and 2nd servo drive is connected to a conveyor plant. Buttons of I_n have also lamb outputs denoted as Q_n . Inputs of I_n are plugged into NX-ID4442 coupler unit and outputs of Q_n are plugged into NX-OD4256 coupler unit. A potentiometer is connected to 1st channel of the NX-DA3603 coupler unit.

The analog output unit is connected to the analog input unit such in a way that the 1st channel of the analog output unit determines the upper limit of the 1st channel of the analog input. This means that if the 1st channel of the analog output unit is *AnalogMax*, the 1st channel of the analog input is in the interval of $[0 \text{ } AnalogMax]$.

¹EtherCAT is an ethernet-based automation protocol used in Omron NJ type CPUs.

2 Create New Project

Sysmac Studio program is used in Omron Sysmac Laboratory to program **NJ** type PLCs. To create a project successfully, follow these steps:

1. Open the program and click *Offline > New Project* to create your project.
2. Fill your project properties which gives information about your project.
3. Select your controller device and its version match with your laboratory set. Your controller may not be the same as given in 1st section. **Check your controller and its version.**
4. After filling all sections, you can create the project.

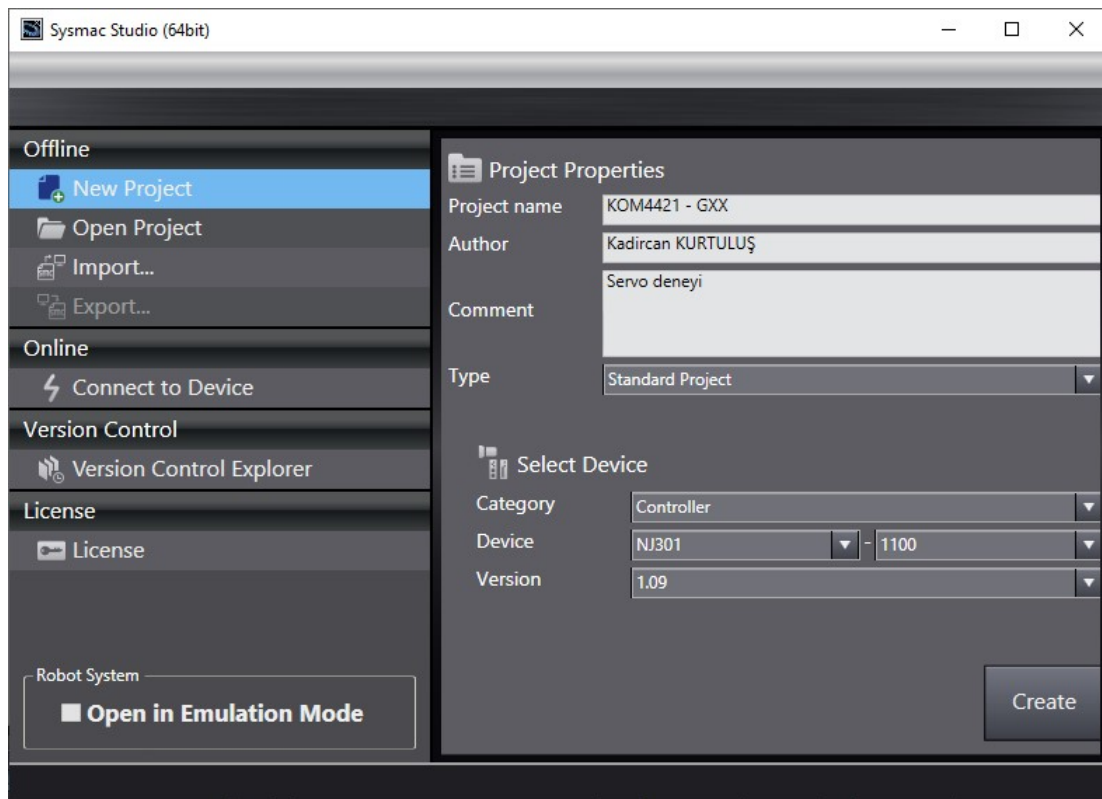


Figure 1: New project screen

3 Configuring Communication Setup

As stated in 1st section; PC, CPU and HMI are connected via an ethernet hub. This must be configured in Sysmac Studio. To achieve this, click *Controller > Communication Setup...* from topbar. From *Communication Setup* window, select *Ethernet connection via a hub*.

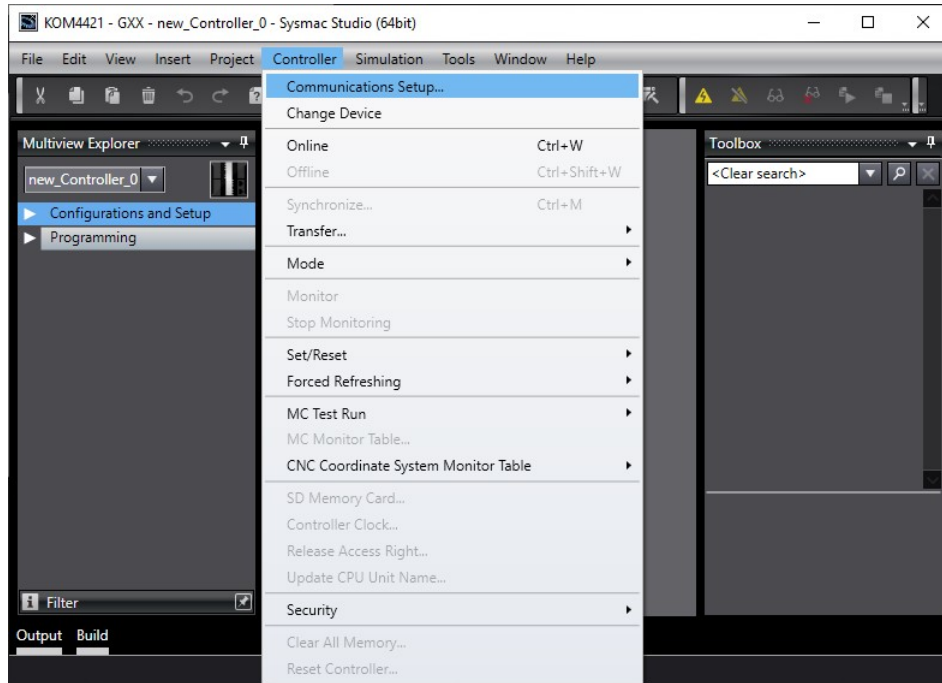


Figure 2: Communication setup

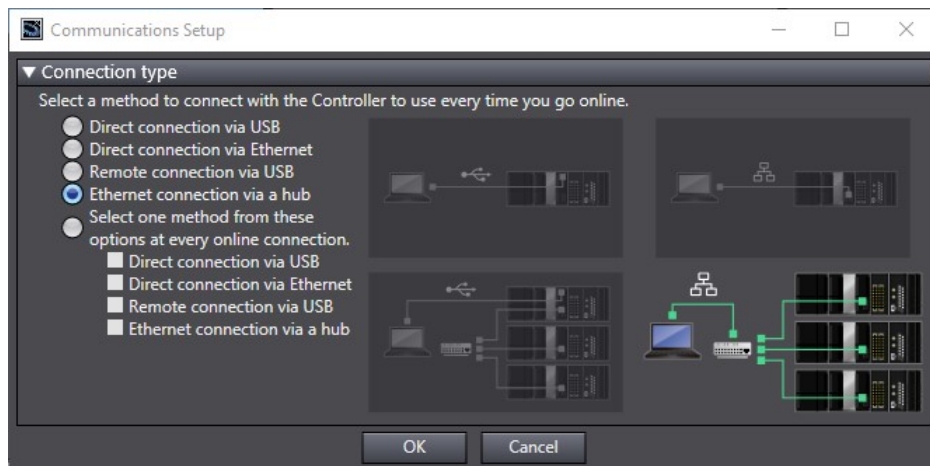


Figure 3: Ethernet connection via a hub

With these settings, you will be able to work online without problems.

4 Adding EtherCAT Slaves

EtherCAT slaves of servo drivers and coupler for your laboratory set must be added to your configuration. To achieve this, click *Configurations and Setup > EtherCAT* from Multiview Explorer.

You can configure (add or remove) your EtherCAT network slaves for your configuration. You must add your EtherCAT slaves as given in 1st section. You can search EtherCAT devices via the textbox in *Toolbox*.

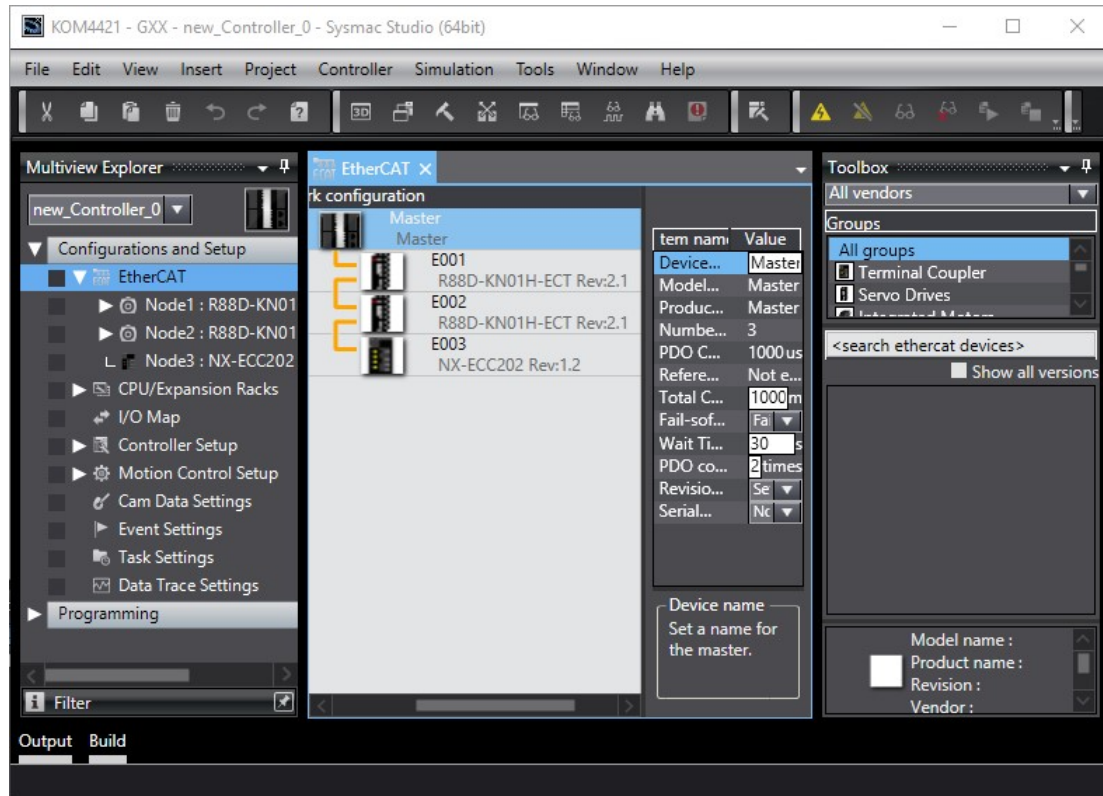


Figure 4: EtherCAT configuration

Note that your EtherCAT slaves may not be the same as given in 1st section. **Check your EtherCAT slaves and their versions.**

5 Adding Coupler Units

The coupler has units to be added to its configuration. Reading and writing data to buttons and analog module is achieved via these units. To configure it, click *Configurations and Setup > EtherCAT > Node 3* from Multiview Explorer.

You can configure (add or remove) coupler units for your configuration. You must add your coupler units as given in 1st section. You can search coupler units via the textbox in Toolbox.

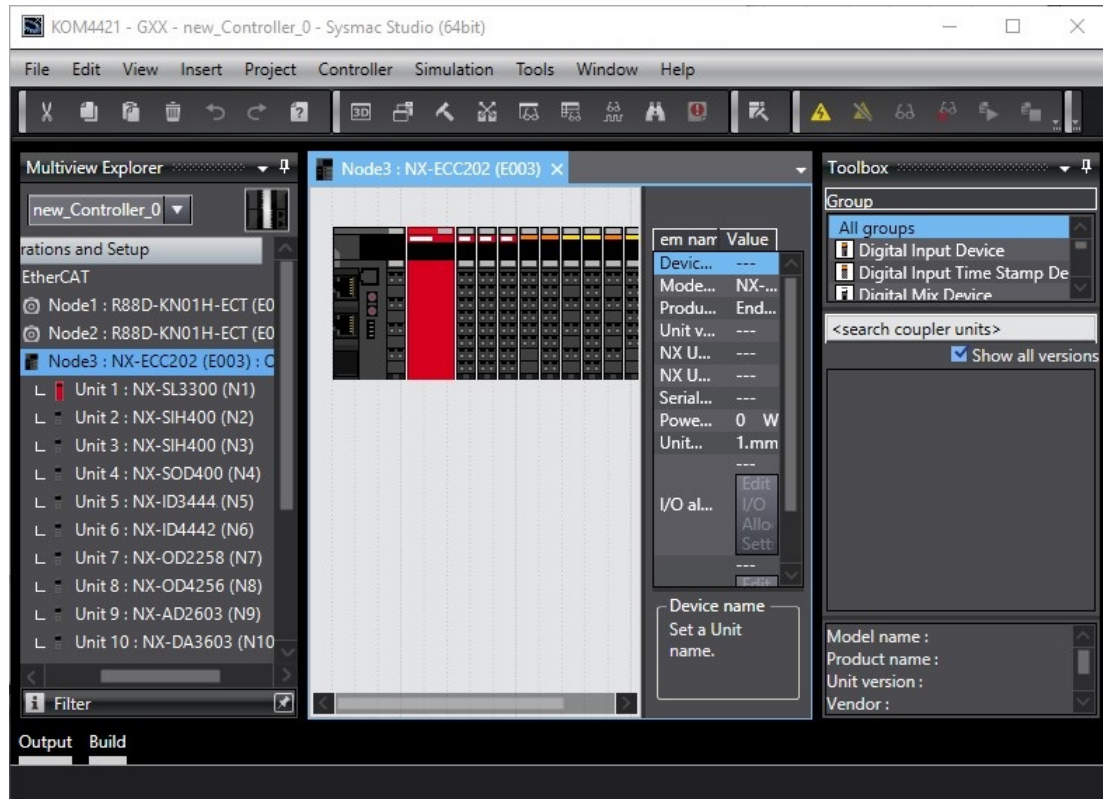


Figure 5: Coupler configuration

Make sure that *Show all versions* is ticked since Safety units are not the latest versions.

Note that your coupler units may not be the same as given in 1st section. Check your coupler units and their versions.

6 Configuring I/O Map

I/O Map is the map which connects ports of the units directly to variables defined locally or globally in the controller. You can connect your buttons and analog values to the variables which are used in programming section. To configure it, click *Configurations and Setup > I/O Map* from Multiview Explorer.

You must make these mappings to NX-ID4442 unit:

0. I_1
1. I_2
2. I_3
3. I_4

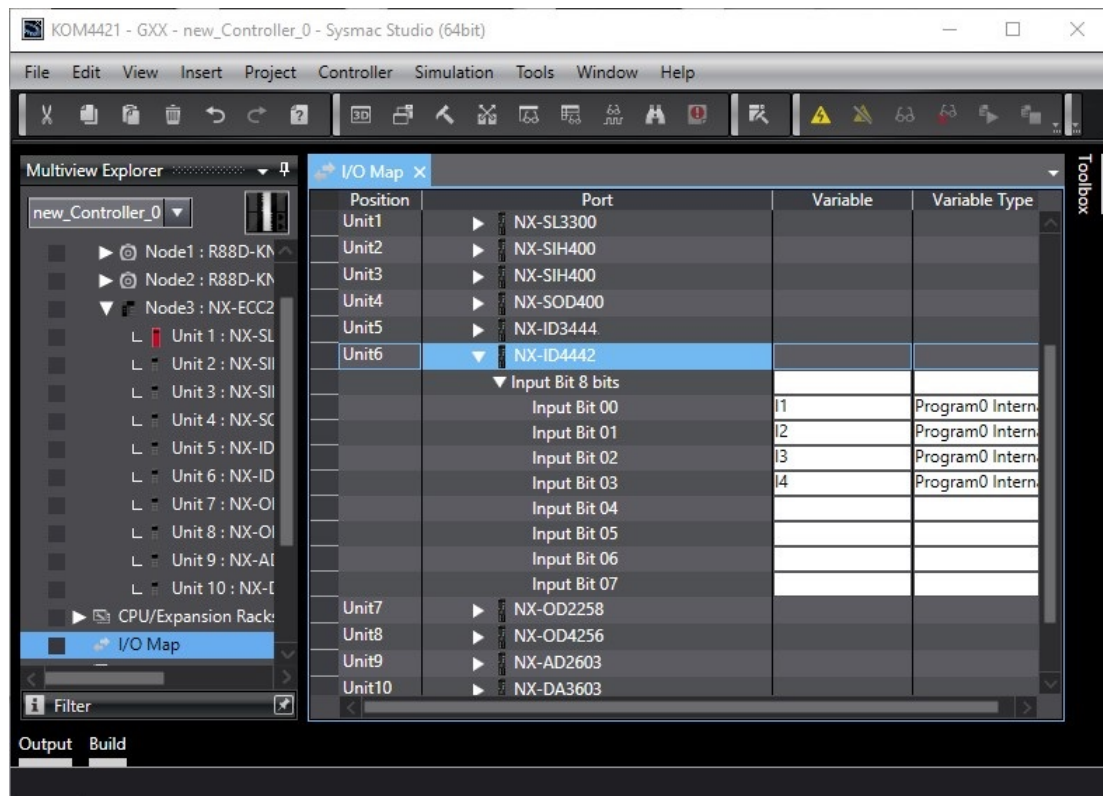


Figure 6: Port configuration of input unit

You must make these mappings to given units below:

- NX-OD4256 Transistor Output Unit
 0. Q_1
 1. Q_2
 2. Q_3
 3. Q_4
- NX-AD2603 Analog Input Unit
 1. AnalogValue
- NX-DA3603 Analog Output Unit
 1. AnalogMax

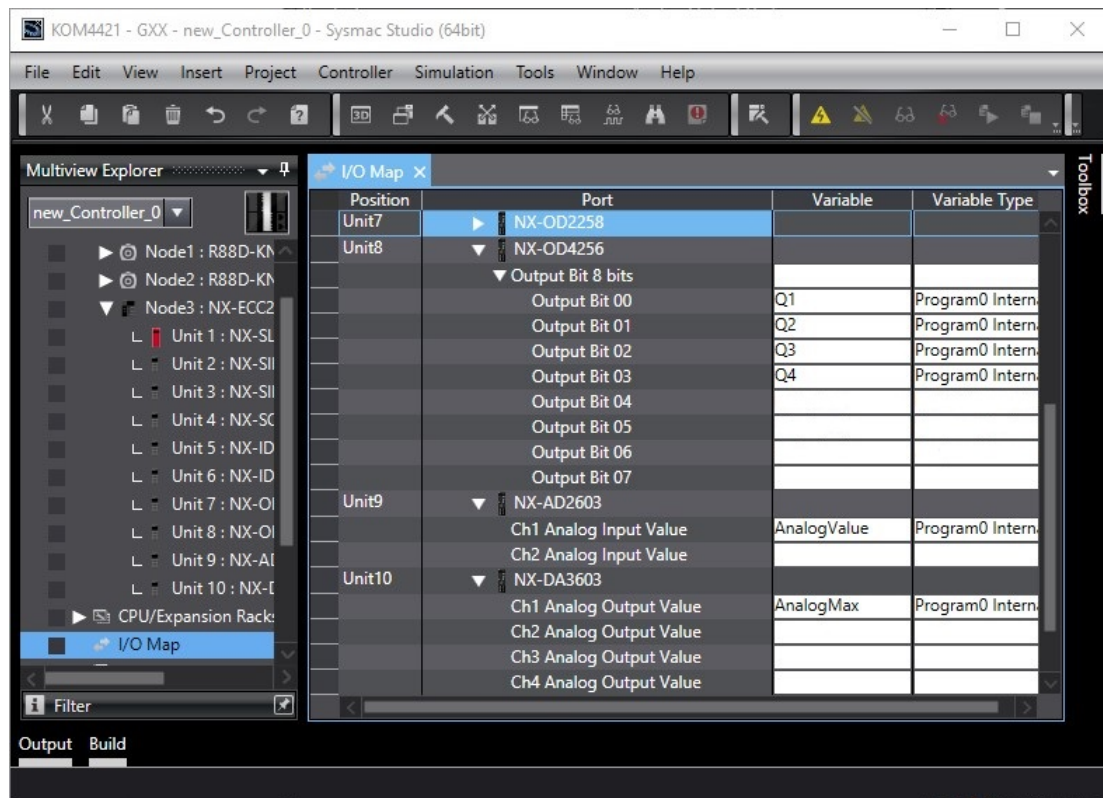


Figure 7: Port configuration of output and 2 analog units

With these mappings, you will map inputs of I_n to variables of I_n and outputs of Q_n to variables of Q_n . You will also assign 1st channel of the analog input, whose upper limit is *AnalogMax*, to *AnalogValue*.

Note that your connections may not be the same as given in 1st section. **Check your connections.**

7 Configuring Motion Control Setup

The first 2 EtherCAT slaves of the controller are servo drives. These slaves must be defined as axes to use in the programming section. To achieve this, click *Configurations and Setup > Motion Control Setup > Axis Settings (right click) > Add > Motion Control Axis* from Multiview Explorer and do this again to get 2 axes. Do not forget to rename your axes as given below. You can rename them by right-clicking and clicking *Rename* from Multiview Explorer.

You must make these settings for the axes given below:

0. RotaryAxis

- Axis Basic Settings
 - Axis type : Servo axis
 - Output device 1 : Node 1
- Unit Conversion Settings
 - Unit of display : degree
 - Command pulse : ?? pulse/rev
 - Work travel distance : 360 degree/rev
- Operation Settings
 - Velocity/Acceleration/Deceleration
 - * Maximum velocity: 100000 degree/s
 - * Maximum jog velocity: 100000 degree/s
- Position Count Settings
 - Count mode : Rotary mode
 - Maximum position : 360 degree
 - Minimum position : 0 degree

1. LinearAxis

- Axis Basic Settings
 - Axis type : Servo axis
 - Output device 1 : Node 2
- Unit Conversion Settings
 - Unit of display : mm
 - Command pulse : ?? pulse/rev
 - Work travel distance : ?? mm/rev
- Operation Settings
 - Velocity/Acceleration/Deceleration
 - * Maximum velocity: 1000 mm/s
 - * Maximum jog velocity: 1000 mm/s
- Position Count Settings
 - Count mode : Linear mode

Both device has a 20-bit encoder. Calculate the pulse/rev value according to this information. The linear axis has a conveyor connected to a wheel with a 16 mm radius. Calculate the mm/rev according to this information.

8 Programming the Controller

After making necessary configurations for EtherCAT, coupler and axes; you are ready to program your controller. You can get to programming section by clicking *Programming > POU's > Programs > Program0 > Section0*. You can search instructions via the textbox in *Toolbox*. You can configure your variables from *Variables* section by clicking up arrow which is located next to the textbox in *Toolbox*.

Before starting to write your algorithm, you must determine the upper limit of the 1st channel of the analog input unit. You can achieve this by assigning *AnalogMax* variable since you have mapped this variable to 1st channel of the analog output unit.

All controller instructions are made via clicking *Controller* from topbar. Before transfer to a Omron controller, you must work online as you have already known.

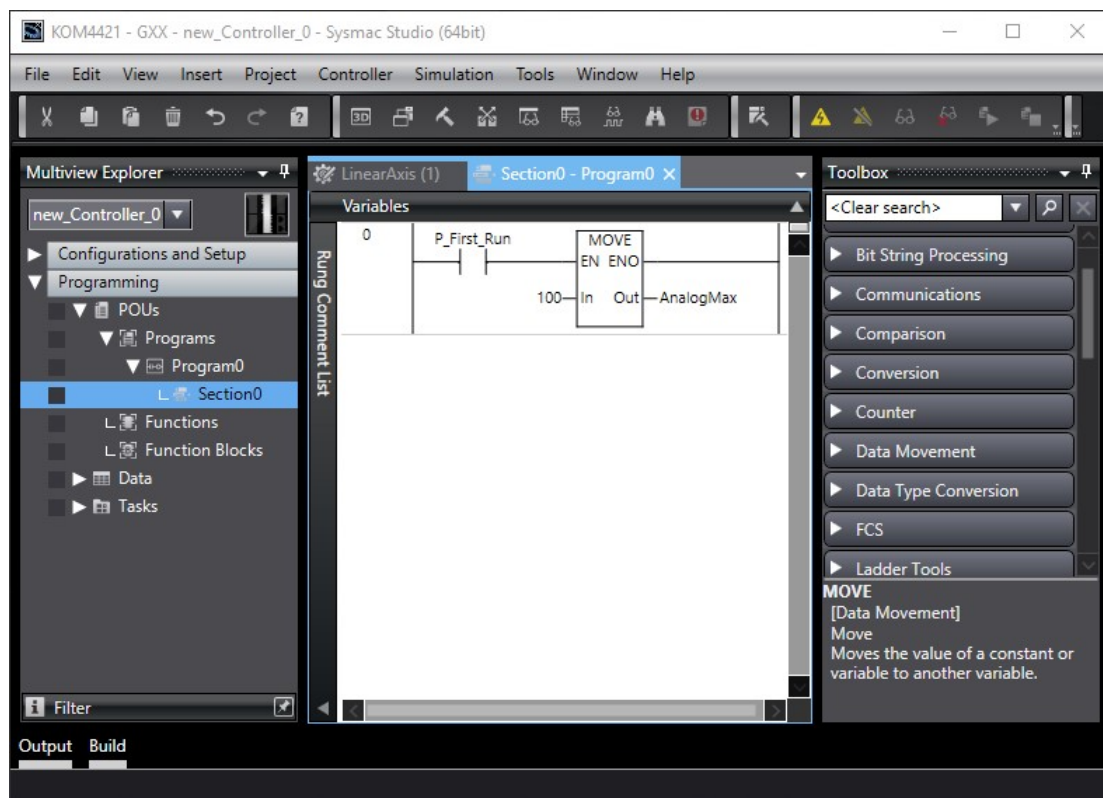


Figure 8: Programming the controller